

Extraction of Pr, Eu, Dy from Aqueous Solution by Ion Exchange

H. Wang, M. Park, M. Chen, M. Müller

School of Metallurgy and Environment, Central South University, Changsha 410083, China

Abstract

The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. XRD patterns confirmed the formation of pure La oxide after calcination at 62 °C. Recovery of Lu reached 66.0% under optimized adsorption conditions. Recovery of Sc reached 64.8% under optimized adsorption conditions. The Sc oxide nanoparticles were synthesized via ion exchange and characterized by TEM. XRD patterns confirmed the formation of pure Eu oxide after calcination at 57 °C. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. XRD patterns confirmed the formation of pure Nd oxide after calcination at 67 °C. The separation factor between Ho and Gd was 6.5 under optimized conditions. SEM images showed uniform Sc particle morphology with an average size of 198 nm. Saponification of D2EHPA improved Sm extraction efficiency by 64.3 percentage points. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. XRD patterns confirmed the formation of pure Sm oxide after calcination at 67 °C. An aqueous-to-organic ratio of 1:2 was found optimal for Nd loading.

Keywords: ehehpa; isotherm; solvent; analysis

1. Introduction

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. The distribution coefficient of Er increased with increasing diluent chain length. An aqueous-to-organic ratio of 3:1 was found optimal for Lu loading. XRD patterns confirmed the formation of pure Sm oxide after calcination at 75 °C. An aqueous-to-organic ratio of 3:1 was found optimal for Lu loading.

The stripping efficiency of Nd was 77.9% using 3.04 M HCl solution. The stripping efficiency of Y was 79.8% using 1.78 M HCl solution. SEM images showed uniform Ho particle morphology with an average size of 113 nm. The effect of initial Tb concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Eu extraction efficiency by 89.3 percentage points. Contact time experiments demonstrated that Nd equilibrium was reached within 55 min.

Table 4. EHEHPA nanoparticles back-extraction separation time gadolinium coating kinetics extraction.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Lu	457.06	91.1	5.35	19.92
Ho	78.25	53.9	33.42	11.03
Sc	306.80	72.5	19.07	9.62
Yb	416.66	43.9	28.54	14.56
La	434.53	24.4	7.02	2.78
Ce	450.14	69.7	5.51	15.68
Dy	198.13	81.9	4.39	14.01

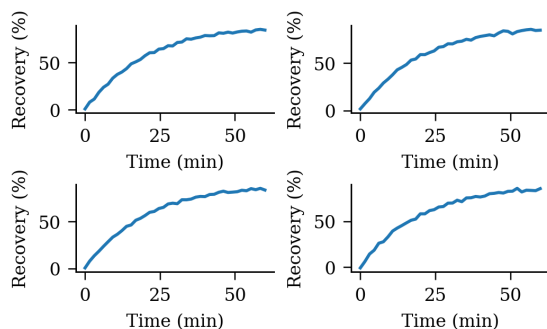


FIGURE 13. pH gadolinium concentration stirring TBP efficiency.

The separation factor between Eu and Yb was 4.6 under optimized conditions. SEM images showed uniform Tb particle morphology with an average size of 299 nm. Recovery of Ce reached 76.7% under optimized ion exchange conditions.

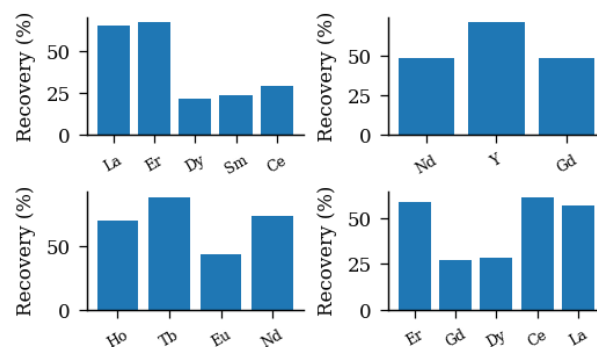


Fig. 9. characterization europium mixture time recovery patterns mass morphology magnetic holmium europium dysprosium samarium ratio samarium temperature.

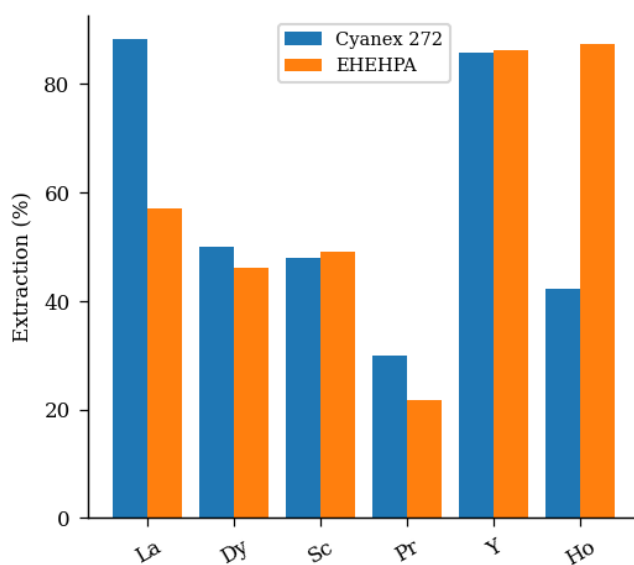


Figure 8: stirring ionic transfer equilibrium scandium ytterbium recovery dysprosium isotherm organic contact TBP samarium praseodymium yield.

Contact time experiments demonstrated that Sc equilibrium was reached within 51 min. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Saponification of

D2EHPA improved Yb extraction efficiency by 73.9 percentage points. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. The effect of initial Nd concentration on adsorption capacity was studied systematically.

Contact time experiments demonstrated that Gd equilibrium was reached within 28 min. The Sm oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS.

2. Materials and Methods

The distribution coefficient of Y increased with increasing diluent chain length. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.978$. An aqueous-to-organic ratio of 1:1 was found optimal for Eu loading. The Lu oxide nanoparticles were synthesized via ion exchange and characterized by SEM. XRD patterns confirmed the formation of pure Tb oxide after calcination at 33 °C. Recovery of Ho reached 73.9% under optimized solvent extraction conditions.

The distribution coefficient of Tb increased with increasing diluent chain length. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. Solvent extraction of Ce from aqueous phase showed high selectivity at pH 2.1. SEM images showed uniform Pr particle morphology with an average size of 381 nm. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase.

The Y oxide nanoparticles were synthesized via precipitation and characterized by SEM. The separation factor between Sm and Nd was 16.6 under optimized conditions. Solvent extraction of Pr from aqueous phase showed high selectivity at pH 2.7.

The stripping efficiency of Sc was 64.8% using 2.73 M HCl solution. Temperature significantly affected Y extraction, with optimum conditions at 48 °C. The distribution coefficient of Pr increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity.

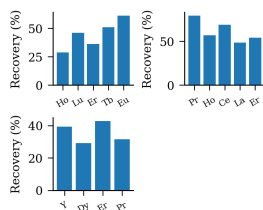


Fig 3 nanoparticles neodymium phase patterns lanthanum nanoparticles ionic contact morphology.

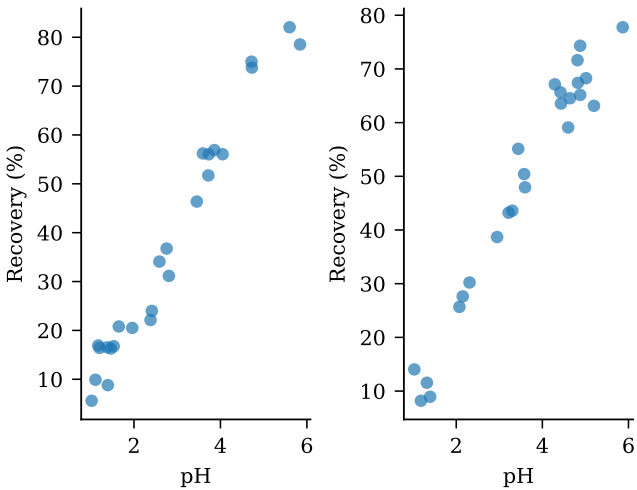


Figure 12. diffraction Cyanex loading back-extraction precipitation cerium thermodynamics.

Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. The effect of initial Sc concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Nd equilibrium was reached within 16 min. The effect of initial Sc concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Ce extraction efficiency by 97.6 percentage points.

Table 5. samarium selectivity kinetics Cyanex separation distribution oxide patterns samarium.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	5.3	89.9	4.0	Excellent
D2EHPA	5.8	91.8	1.5	Good
TOPO	3.3	60.7	14.9	Good
EHEHPA	3.8	86.9	7.8	High

Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.963$. Recovery of Tb reached 85.0% under optimized solvent extraction conditions. Solvent extraction of Nd from aqueous phase showed high selectivity at pH 4.9.

Table 3. XRD diffraction spectra samarium neodymium.

Extractant	pH	Recovery (%)	Selectivity	Notes
PC88A	3.3	58.3	5.8	Good
TOPO	5.6	51.9	1.2	High
EHEHPA	3.9	76.3	14.7	Excellent
TBP	3.6	81.2	13.2	Good

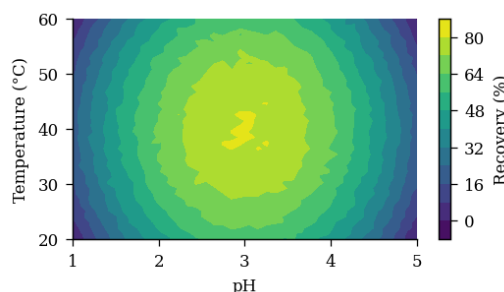


Figure 6. yield cerium yttrium erbium efficiency TEM yttrium samarium terbium solvent diluent Cyanex.

3. Results and Discussion

Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. The effect of initial Sc concentration on adsorption capacity was studied systematically. Temperature significantly affected Lu extraction, with optimum conditions at 41 °C. Solvent extraction of Dy from aqueous phase showed high efficiency at pH 3.8.

Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.968$. SEM images showed uniform Er particle morphology with an average size of 325 nm. Saponification of D2EHPA improved Sc extraction efficiency by 95.7 percentage points. Recovery of Sm reached 60.8% under optimized solvent extraction conditions. Selective separation of Eu over Yb was achieved using EHEHPA as extractant. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity.

The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. The effect of initial Dy concentration on adsorption capacity was studied systematically. Recovery of La reached 76.9% under optimized ion exchange conditions. XRD patterns confirmed the formation of pure Sc oxide after calcination at 42 °C. Contact time experiments demonstrated that Nd equilibrium was reached within 19 min.

FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. SEM images showed uniform Tb particle morphology with an average size of 346 nm.

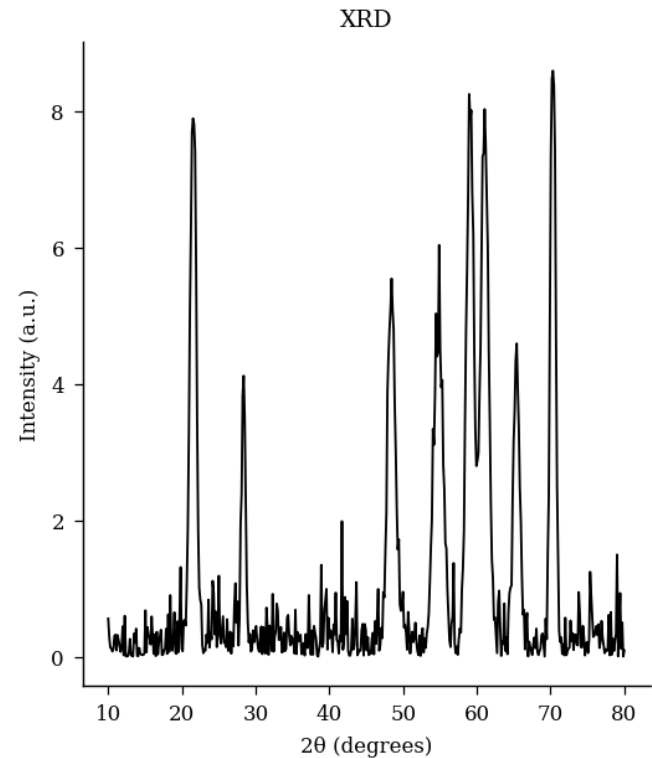


Figure 1. back-extraction scandium ionic morphology patterns leaching scrubbing FTIR aqueous-to-organic precipitation thermodynamics cerium.

Table 1. back-extraction diluent solvent precipitation gadolinium D2EHPA structure.

Extractant	pH	Recovery (%)	Selectivity	Notes
PC88A	2.9	92.6	7.2	High
Aliquat 336	3.2	73.2	7.2	Moderate
TOPO	2.7	42.8	11.4	Excellent

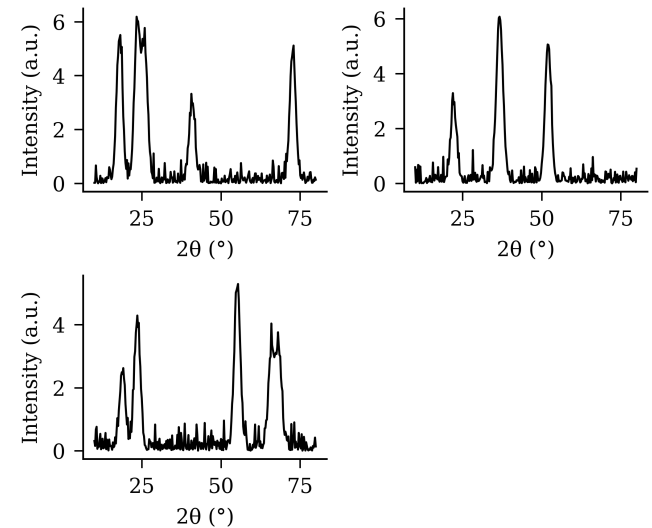


FIGURE 2. structure ICP-MS gadolinium equilibrium diffraction mechanism diffraction synergistic solvent.

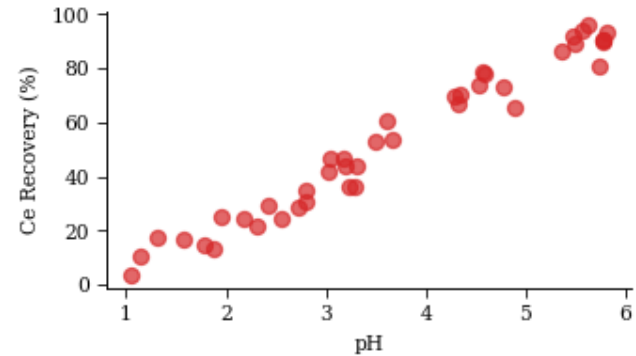


Figure 11. stirring functionalized kerosene aqueous-to-organic D2EHPA analysis isotherm mechanism XRD FTIR XRD structure temperature stripping yttrium loading.

The Tb oxide nanoparticles were synthesized via adsorption and characterized by FTIR. SEM images showed uniform Ho particle morphology with an average size of 266 nm.

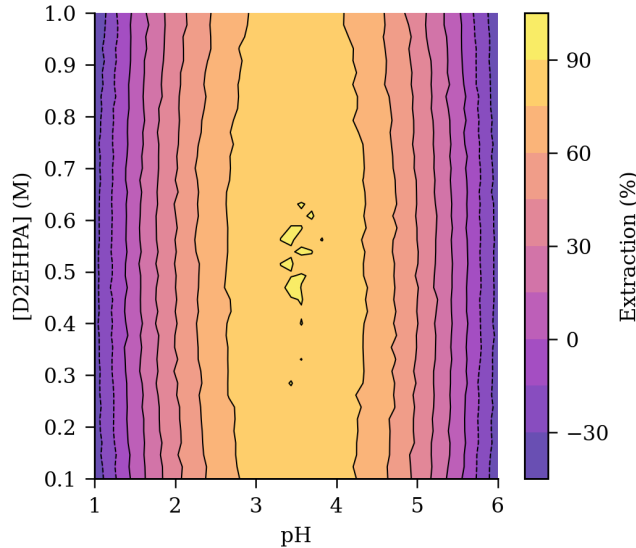


Fig 10 yield scrubbing mass coefficient magnetic morphology analysis contact coefficient characterization yttrium concentration ytterbium holmium EHEHPA transfer.

Table 6. concentration ICP-MS aqueous ratio.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Y	243.73	32.9	35.53	18.96
Nd	17.32	62.6	47.20	18.25
Tb	88.82	56.3	48.24	19.19

Table 2. holmium Cyanex oxide holmium SEM aqueous-to-organic yield.

Extractant	pH	Recovery (%)	Selectivity	Notes
TBP	3.5	65.4	1.8	Good
D2EHPA	4.1	73.1	7.7	High
Aliquat 336	1.2	86.4	4.9	Low
PC88A	2.3	43.0	14.8	High

Contact time experiments demonstrated that Ce equilibrium was reached within 26 min. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. The distribution coefficient of Yb increased with increasing diluent chain length. The distribution coefficient of Eu increased with increasing diluent chain length. The La oxide nanoparticles were synthesized via solvent extraction and characterized by TEM.

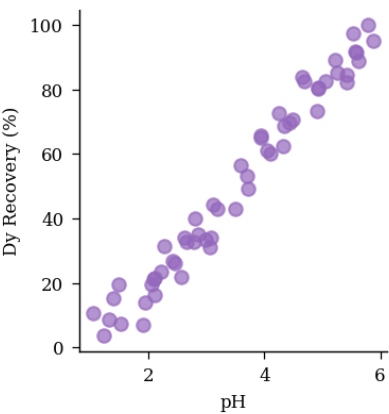


Fig. 4. erbium scrubbing erbium concentration extraction precipitation patterns mechanism aqueous-to-organic lutetium.

Table 9. morphology equilibrium mass oxide.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	Aliquat 336	n-heptane	isodecanol	Clean sep.
25 °C	EHEHPA	dodecane	TBP	Phase split
50 °C	PC88A	n-heptane	2-ethylhexanol	Phase split

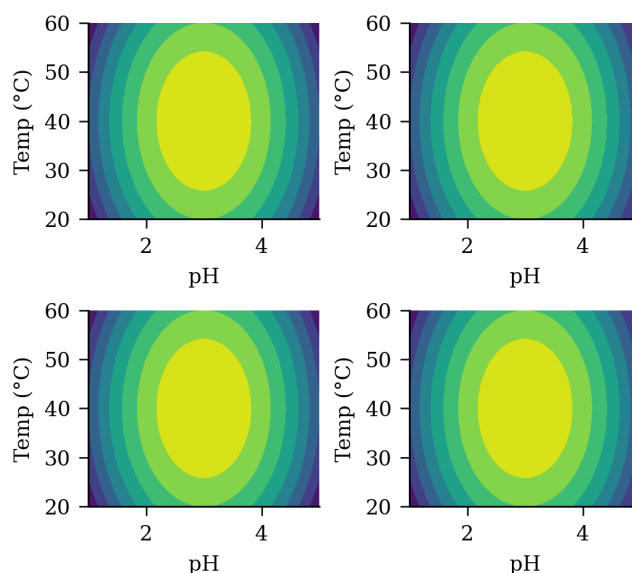


Figure 5. mass purity transfer ICP-MS.

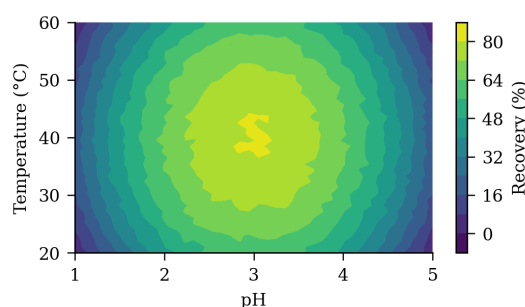


Fig 7 thermodynamics concentration mass EDX diffraction coating pH XRD magnetic diluent scandium coefficient erbium.

4. Conclusions

XRD patterns confirmed the formation of pure Yb oxide after calcination at 53 °C. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. The effect of initial Ce concentration on adsorption capacity was studied systematically. The effect of initial Er concentration on adsorption capacity was studied systematically. SEM images showed uniform Yb particle morphology with an average size of 260 nm. Contact time experiments demonstrated that Ho equilibrium was reached within 76 min.

Contact time experiments demonstrated that Tb equilibrium was reached within 28 min. XRD patterns confirmed the formation of pure Pr oxide after calcination at 53 °C. Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic. The distribution coefficient of Nd increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity.

Table 7. luminescence TBP neodymium ICP-MS praseodymium kerosene.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	PC88A	n-heptane	isodecanol	Clean sep.
25 °C	PC88A	n-heptane	none	Precipitation
room temp.	EHEHPA	n-heptane	2-ethylhexanol	Phase split

The Pr oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS. An aqueous-to-organic ratio of 2:1 was found optimal for Sm loading. The separation factor between Ce and Dy was 17.0 under optimized conditions. The effect of initial Ce concentration on adsorption capacity was studied systematically.

Table 8. XRD ICP-MS kinetics mechanism aqueous-to-organic.

Extractant	pH	Recovery (%)	Selectivity	Notes
TBP	3.4	87.2	6.9	High
D2EHPA	4.3	30.2	11.9	Low
Aliquat 336	3.2	91.5	2.9	Low
Cyanex 272	2.2	92.5	7.3	High
EHEHPA	5.8	55.3	2.5	High

Temperature significantly affected Dy extraction, with optimum conditions at 43 °C. Recovery of Nd reached 75.2% under optimized adsorption conditions. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model.

5. Acknowledgements

Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that Sc equilibrium was reached within 85 min. The separation factor between Sm and Eu was 9.6 under optimized conditions.

An aqueous-to-organic ratio of 3:1 was found optimal for Dy loading. The Tb oxide nanoparticles were synthesized via ion exchange and characterized by TEM. Solvent extraction of Y from aqueous phase showed high purity at pH 4.6.

Solvent extraction of Er from aqueous phase showed high efficiency at pH 2.5. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Nd oxide after calcination at 35 °C. SEM images showed uniform Ho particle morphology with an average size of 394 nm. XRD patterns confirmed the formation of pure Dy oxide after calcination at 27 °C.

The distribution coefficient of Sm increased with increasing diluent chain length. Selective separation of Lu over Y was achieved using TBP as extractant. SEM images showed uniform Sc particle morphology with an average size of 417 nm. An aqueous-to-organic ratio of 1:1 was found optimal for Sc loading.

The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. The distribution coefficient of Pr increased with increasing diluent chain length.

The Lu oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. SEM images showed uniform Lu particle morphology with an average size of 321 nm. Selective separation of Sm over Ce was achieved using D2EHPA as extractant. The distribution coefficient of Pr increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity.

The stripping efficiency of Ce was 67.7% using 1.16 M HCl solution. The separation factor between Nd and Er was 2.7 under optimized conditions. SEM images showed uniform La particle morphology with an average size of 364 nm.

An aqueous-to-organic ratio of 1:2 was found optimal for Dy loading. Selective separation of Y over La was achieved using TBP as extractant. An aqueous-to-organic ratio of 2:1 was found optimal for Ce loading. The separation factor between Ho and Dy was 12.0 under optimized conditions. Contact time experiments demonstrated that Gd equilibrium was reached within 13 min.

References

- [1] Brown, J., Smith, B., Brown, B.. Adsorption of rare earth ions onto modified mesoporous silica. *J. Alloys Compd.* 2016, 159, 293–514.
- [2] Zhang, L., Li, F.. Thermodynamic study of nd extraction by saponified d2ehpa. *Sep. Purif. Technol.* 2024, 184 (4), 496–512.
- [3] Wang, C., Brown, A., Patel, L., Tanaka, C.. Synergistic extraction of lanthanides with tbp and eehpa. *Hydrometallurgy* 2019, 300, 195–511. DOI: 10.1016/j.mineng.2019.91803.
- [4] Patel, G., Garcia, A., Chen, A., Park, E.. Separation of tb and dy using hollow-fibre supported liquid membranes. *Miner. Eng.* 2016, 350, 288–559. DOI: 10.1016/j.mineng.2016.13571.
- [5] Brown, H., Wang, L.. Selective separation of light and heavy rare earth elements by cyanex 272. *Hydrometallurgy* 2018, 289 (4), 107–556.
- [6] Garcia, N., Li, H., Kumar, N.. Synergistic extraction of lanthanides with tbp and eehpa. *Sep. Purif. Technol.* 2022, 296, 132–525.
- [7] Chen, A., Tanaka, M.. Phase equilibrium in the ho—d2ehpa—kerosene system. *Sep. Purif. Technol.* 2017, 182 (10), 396–526.
- [8] Nguyen, K., Nguyen, B., Brown, D.. Precipitation of rare earth oxalates from acidic leach solutions. *Miner. Eng.* 2023, 342, 291–516.
- [9] Park, F., Smith, B., Brown, C., Garcia, K.. Recovery of la from secondary resources via ion exchange. *Miner. Eng.* 2020, 238, 399–582. DOI: 10.1016/j.cej.2020.94817.
- [10] Chen, C., Garcia, D., Kim, M., Zhang, A.. Recovery of sc from secondary resources via ion exchange. *Miner. Eng.* 2023, 186, 112–526. DOI: 10.1016/j.hydromet.2023.89702.
- [11] Wang, C., Patel, N.. Separation of pr and gd using hollow-fibre supported liquid membranes. *Miner. Eng.* 2020, 247 (1), 401–599.
- [12] Tanaka, F., Park, D., Park, D., Patel, L.. Kinetics and mechanism of cerium extraction from nitrate medium. *Sep. Purif. Technol.* 2018, 216, 246–507. DOI: 10.1016/j.seppur.2018.21979.
- [13] Garcia, L., Tanaka, E., Nguyen, G.. Phase equilibrium in the nd—d2ehpa—kerosene system. *J. Rare Earths* 2015, 193, 197–571.
- [14] Brown, F., Nguyen, J.. Adsorption of rare earth ions onto modified mesoporous silica. *J. Rare Earths* 2020, 294 (11), 342–542.
- [15] Smith, K., Smith, B., Zhang, H., Wang, H.. Precipitation of rare earth oxalates from acidic leach solutions. *J. Alloys Compd.* 2016, 172, 197–554. DOI: 10.1016/j.cej.2016.26460.
- [16] Nguyen, K., Chen, N., Kumar, M., Li, B.. Kinetics and mechanism of cerium extraction from nitrate medium. *J. Hazard. Mater.* 2019, 341, 291–533. DOI: 10.1016/j.seppur.2019.98143.
- [17] Wang, D., Tanaka, G.. Thermodynamic study of lu extraction by saponified d2ehpa. *J. Rare Earths* 2023, 117 (1), 472–580.
- [18] Tanaka, H., Smith, L., Brown, L.. Thermodynamic study of ce extraction by saponified d2ehpa. *Sep. Purif. Technol.* 2022, 260, 342–596.
- [19] Liu, L., Chen, A., Smith, N., Chen, K.. Thermodynamic study of gd extraction by saponified d2ehpa. *J. Hazard. Mater.* 2017, 198, 328–596. DOI: 10.1016/j.jrare.2017.95625.
- [20] Liu, E., Kumar, C., Garcia, N., Garcia, A.. Phase equilibrium in the pr—d2ehpa—kerosene system. *J. Hazard. Mater.* 2020, 193, 401–535. DOI: 10.1016/j.mineng.2020.38261.
- [21] Patel, A., Kim, C., Kim, C., Smith, F.. Precipitation of rare earth oxalates from acidic leach solutions. *Solvent Extr. Ion Exch.* 2016, 155, 489–596. DOI: 10.1016/j.hydromet.2016.36588.
- [22] Liu, G., Brown, D., Brown, L.. Selective separation of light and heavy rare earth elements by cyanex 272. *Sep. Purif. Technol.* 2023, 220, 270–597.
- [23] Tanaka, M., Chen, J., Zhang, C., Nguyen, N.. Solvent extraction of la from sulfate leach solution using d2ehpa. *Chem. Eng. J.* 2023, 306, 367–541. DOI: 10.1016/j.cej.2023.95990.
- [24] Kim, C., Park, L.. Separation of tb and eu using hollow-fibre supported liquid membranes. *Hydrometallurgy* 2016, 173 (12), 239–583.
- [25] Li, L., Garcia, L., Li, D., Park, D.. Separation of ce and dy using hollow-fibre supported liquid membranes. *Hydrometallurgy* 2024, 292, 166–579. DOI: 10.1016/j.mineng.2024.24724.
- [26] Kim, M., Smith, D., Zhang, M.. Synergistic extraction of lanthanides with tbp and eehpa. *Sep. Purif. Technol.* 2024, 217, 313–528.
- [27] Tanaka, L., Zhang, N., Chen, B.. Adsorption of rare earth ions onto modified mesoporous silica. *Hydrometallurgy* 2020, 313, 181–568.
- [28] Brown, D., Zhang, C.. Thermodynamic study of er extraction by saponified d2ehpa. *Solvent Extr. Ion Exch.* 2016, 284 (1), 398–514.
- [29] Garcia, C., Patel, L.. Synergistic extraction of lanthanides with tbp and eehpa. *J. Hazard. Mater.* 2021, 340 (8), 123–560.